Machuzu FHIR

28 March 2024

Presentation 1

Introduction

As a patient your medical data is collected on many systems, and at different national and international locations. The IPA specification is designed to help patients access their data regardless of the system on which it was captured or the location of the patient. This is particularly useful in several scenarios such as;

- when you are travelling out of your home country and need your medical data while away
- when you are unconscious and cannot provide your medical history
- when you are at a local health facility using a different system than that of your regular facility

The system's main purpose is to be able to consolidate and share a person's health data. This is achieved by the system being implemented in FHIR IPA standards. These allow for compatible FHIR systems to exchange data with the system.

Current Status

IPA is not meant to be a complete Electronic Medical Record system, but instead it captures the most crucial elements of the EMR or pulls the required data from other systems. However, due to that it is mainly developed for interoperability; it is able to pull and link required medical data from other systems.

Ongoing Development

There is further functionality that is pending for the system to be complete and among the planned activities for development is;

- Financial / Billing module; enabling payments to be made and linking to medical aid schemes
- Telemedicine module; enabling patients to get assistance online
- A ***Smart Health Card***; a secure smart card containing a patient's health records implementing Blockchain technology and the ISO/IEEE 11073 standard https://build.fhir.org/ig/HL7/fhir-shc-vaccination-ig/ , https://smarthealth.cards/en/ , https://smarthealth.cards/en/how-they-work.html , https://spec.smarthealth.cards/

Smart Card Examples

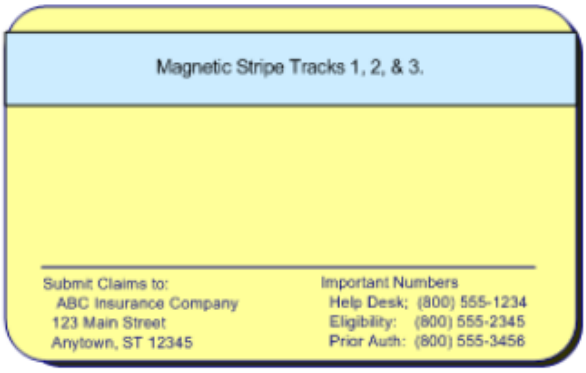
Provided by Health Facilities. Data accessed via this card includes ID Number, Barcode or QRCode and is limited to Personal Details, Vaccination and Immunisation data.





Provided by Us (Provider). Our cards have chips or small microcontrollers instead of Magnetic Stripe, which was used in the olden days for Bank and other smart cards. Data accessed via this card is 100% secure and can only be accessed via a compatible Smart Card Reader that needs to be connected to a computer running our software. Data that can be accessed without security can be retrieved using the ID Number and is limited to Personal Details, Vaccination and Immunisation data.





The cost of smart cards is very low and ranges from 4 – 10 USD. Compatible Smart Card readers can be purchased for around 40 USD.

Although as illustrated; we shall not only provide secure smart cards, but also regular cards which shall be printed with a barcode or QRCode.

Presentation Objectives

The objective of the presentation is to;

- Establish if the parameters that have been added are sufficient to run a private practise
- Discuss application lists such as Medicines, Diagnosis, Conditions etc..

- Establish if there is market for the product
- Collect additional requirements for IPA
- Collect additional requirements for private practises
- Establish market trends and practises
- Establish interested implementing or business partners

Parameters

- Basic patient details (shall later be converted to Blockchain)

Name	Flags	Card.	Type	Description & Constraints
Patient	N		DomainResource	Information about an individual or animal receiving health care services Elements defined in Ancestors: id, meta, implicitRules, language, text, contained, extension, modifierExtension
identifier	Σ	0..*	Identifier	An identifier for this patient
active	?! Σ	0..1	boolean	Whether this patient's record is in active use
name	Σ	0..*	HumanName	A name associated with the patient
telecom	Σ	0..*	ContactPoint	A contact detail for the individual
gender	Σ	0..1	code	male | female | other | unknown AdministrativeGender (Required)
birthDate	Σ	0..1	date	The date of birth for the individual
deceased[x]	?! Σ	0..1		Indicates if the individual is deceased or not
deceasedBoolean			boolean	
deceasedDateTime			dateTime	
address	Σ	0..*	Address	An address for the individual
maritalStatus		0..1	CodeableConcept	Marital (civil) status of a patient MaritalStatus (Extensible)
multipleBirth[x]		0..1		Whether patient is part of a multiple birth
multipleBirthBoolean			boolean	
multipleBirthInteger			integer	
photo		0..*	Attachment	Image of the patient
contact	I	0..*	BackboneElement	A contact party (e.g. guardian, partner, friend) for the patient *+ Rule: SHALL at least contain a contact's details or a reference to an organization*
relationship		0..*	CodeableConcept	The kind of relationship Patient Contact Relationship (Extensible)
name		0..1	HumanName	A name associated with the contact person
telecom		0..*	ContactPoint	A contact detail for the person
address		0..1	Address	Address for the contact person
gender		0..1	code	male | female | other | unknown AdministrativeGender (Required)
organization	I	0..1	Reference(Organization)	Organization that is associated with the contact
period		0..1	Period	The period during which this contact person or organization is valid to be contacted relating to this patient
communication		0..*	BackboneElement	A language which may be used to communicate with the patient about his or her health
language		1..1	CodeableConcept	The language which can be used to communicate with the patient about his or her health Common Languages (Preferred but limited to AllLanguages)
preferred		0..1	boolean	Language preference indicator
generalPractitioner		0..*	Reference(Organization | Practitioner | PractitionerRole)	Patient's nominated primary care provider
managingOrganization	Σ	0..1	Reference(Organization)	Organization that is the custodian of the patient record
link	?! Σ	0..*	BackboneElement	Link to another patient resource that concerns the same actual person
other	Σ	1..1	Reference(Patient | RelatedPerson)	The other patient or related person resource that the link refers to
type	Σ	1..1	code	replaced-by | replaces | refer | seealso LinkType (Required)

Documentation for this format

- Problems / Conditions

Name	Flags	Card.	Type	Description & Constraints
Condition	I TU		DomainResource	Detailed information about conditions, problems or diagnoses *+ Guideline: Condition.clinicalStatus SHALL be present if verificationStatus is not entered-in-error and category is problem-list-item* *+ Rule: If condition is abated, then clinicalStatus must be either inactive, resolved, or remission* *+ Rule: Condition.clinicalStatus SHALL NOT be present if verification Status is entered-in-error* Elements defined in Ancestors: id, meta, implicitRules, language, text, contained, extension, modifierExtension
identifier	Σ	0..*	Identifier	External Ids for this condition
clinicalStatus	?! Σ I	0..1	CodeableConcept	active | recurrence | relapse | inactive | remission | resolved Condition Clinical Status Codes (Required)
verificationStatus	?! Σ I	0..1	CodeableConcept	unconfirmed | provisional | differential | confirmed | refuted | entered-in-error ConditionVerificationStatus (Required)
category		0..*	CodeableConcept	problem-list-item | encounter-diagnosis Condition Category Codes (Extensible)
severity		0..1	CodeableConcept	Subjective severity of condition Condition/Diagnosis Severity (Preferred)
code	Σ	0..1	CodeableConcept	Identification of the condition, problem or diagnosis Condition/Problem/Diagnosis Codes (Example)
bodySite	Σ	0..*	CodeableConcept	Anatomical location, if relevant SNOMED CT Body Structures (Example)
subject	Σ	1..1	Reference(Patient | Group)	Who has the condition?
encounter	Σ	0..1	Reference(Encounter)	Encounter created as part of
onset[x]	Σ	0..1		Estimated or actual date, date-time, or age
onsetDateTime			dateTime	
onsetAge			Age	
onsetPeriod			Period	
onsetRange			Range	
onsetString			string	
abatement[x]	I	0..1		When in resolution/remission
abatementDateTime			dateTime	
abatementAge			Age	
abatementPeriod			Period	
abatementRange			Range	
abatementString			string	
recordedDate	Σ	0..1	dateTime	Date record was first recorded
recorder	Σ	0..1	Reference(Practitioner | PractitionerRole | Patient | RelatedPerson)	Who recorded the condition
~~asserter~~	~~Σ~~	~~0..1~~	~~Reference(Practitioner | PractitionerRole | Patient | RelatedPerson)~~	~~Person who asserts this condition~~
~~stage~~	~~I~~	~~0..*~~	~~BackboneElement~~	~~Stage/grade, usually assessed formally~~ *+ Rule: Stage SHALL have summary or assessment*
summary	I	0..1	CodeableConcept	Simple summary (disease specific) Condition Stage (Example)
assessment	I	0..*	Reference(ClinicalImpression | DiagnosticReport | Observation)	Formal record of assessment
type		0..1	CodeableConcept	Kind of staging Condition Stage Type (Example)
~~evidence~~	~~I~~	~~0..*~~	~~BackboneElement~~	~~Supporting evidence~~ *+ Rule: evidence SHALL have code or details*
code	Σ I	0..*	CodeableConcept	Manifestation/symptom Manifestation and Symptom Codes (Example)
detail	Σ I	0..*	Reference(Any)	Supporting information found elsewhere
note		0..*	Annotation	Additional information about the Condition

Lists

- https://hl7.org/fhir/R4/valueset-condition-code.html
- https://hl7.org/fhir/R4/valueset-body-site.html

- Medication orders and usage

Name	Flags	Card.	Type	Description & Constraints
MedicationRequest	TU		DomainResource	Ordering of medication for patient or group Elements defined in Ancestors: id, meta, implicitRules, language, text, contained, extension, modifierExtension
identifier		0..*	Identifier	External ids for this request
status	?! Σ	1..1	code	active | on-hold | cancelled | completed | entered-in-error | stopped | draft | unknown Medicationrequest status (Required)
statusReason		0..1	CodeableConcept	Reason for current status Medication request status reason codes (Example)
intent	?! Σ	1..1	code	proposal | plan | order | original-order | reflex-order | filler-order | instance-order | option Medication request intent (Required)
category		0..*	CodeableConcept	Type of medication usage Medication request category codes (Example)
priority	Σ	0..1	code	routine | urgent | asap | stat Request priority (Required)
doNotPerform	?! Σ	0..1	boolean	True if request is prohibiting action
reported[x]	Σ	0..1		Reported rather than primary record
reportedBoolean			boolean	
reportedReference			Reference(Patient | Practitioner | PractitionerRole | RelatedPerson | Organization)	
medication[x]	Σ	1..1		Medication to be taken SNOMED CT Medication Codes (Example)
medicationCodeableConcept			CodeableConcept	
medicationReference			Reference(Medication)	
subject	Σ	1..1	Reference(Patient | Group)	Who or group medication request is for
encounter		0..1	Reference(Encounter)	Encounter created as part of encounter/admission/stay
supportingInformation		0..*	Reference(Any)	Information to support ordering of the medication
authoredOn	Σ	0..1	dateTime	When request was initially authored
requester	Σ	0..1	Reference(Practitioner | PractitionerRole | Organization | Patient | RelatedPerson | Device)	Who/What requested the Request
performer		0..1	Reference(Practitioner | PractitionerRole | Organization | Patient | Device | RelatedPerson | CareTeam)	Intended performer of administration
performerType	Σ	0..1	CodeableConcept	Desired kind of performer of the medication administration Procedure Performer Role Codes (Example)
recorder		0..1	Reference(Practitioner | PractitionerRole)	Person who entered the request
reasonCode		0..*	CodeableConcept	Reason or indication for ordering or not ordering the medication Condition/Problem/Diagnosis Codes (Example)
reasonReference		0..*	Reference(Condition | Observation)	Condition or observation that supports why the prescription is being written
instantiatesCanonical	Σ	0..*	canonical()	Instantiates FHIR protocol or definition
instantiatesUri	Σ	0..*	uri	Instantiates external protocol or definition
basedOn	Σ	0..*	Reference(CarePlan | MedicationRequest | ServiceRequest | ImmunizationRecommendation)	What request fulfills
groupIdentifier	Σ	0..1	Identifier	Composite request this is part of
courseOfTherapyType		0..1	CodeableConcept	Overall pattern of medication administration Medication request course of therapy codes (Example)
insurance		0..*	Reference(Coverage | ClaimResponse)	Associated insurance coverage
note		0..*	Annotation	Information about the prescription
dosageInstruction		0..*	Dosage	How the medication should be taken
dispenseRequest		0..1	BackboneElement	Medication supply authorization
initialFill		0..1	BackboneElement	First fill details
quantity		0..1	SimpleQuantity	First fill quantity
duration		0..1	Duration	First fill duration
dispenseInterval		0..1	Duration	Minimum period of time between dispenses
validityPeriod		0..1	Period	Time period supply is authorized for
numberOfRepeatsAllowed		0..1	unsignedInt	Number of refills authorized
quantity		0..1	SimpleQuantity	Amount of medication to supply per dispense
expectedSupplyDuration		0..1	Duration	Number of days supply per dispense
performer		0..1	Reference(Organization)	Intended dispenser
substitution		0..1	BackboneElement	Any restrictions on medication substitution
allowed[x]		1..1		Whether substitution is allowed or not V3 Value SetActSubstanceAdminSubstitutionCode (Example)
allowedBoolean			boolean	
allowedCodeableConcept			CodeableConcept	
reason		0..1	CodeableConcept	Why should (not) substitution be made V3 Value SetSubstanceAdminSubstitutionReason (Example)
priorPrescription		0..1	Reference(MedicationRequest)	An order/prescription that is being replaced
detectedIssue		0..*	Reference(DetectedIssue)	Clinical Issue with action
eventHistory		0..*	Reference(Provenance)	A list of events of interest in the lifecycle

NB: dosageInstruction was replaced by Medication due to the logic that dosages are assigned to particular medicines. Also look into MedicationStatement. I am not sure what it is for @ https://build.fhir.org/ig/HL7/fhir-ipa/StructureDefinition-ipa-medicationstatement.html

Lists

- https://hl7.org/fhir/R4/medication.html
- https://hl7.org/fhir/R4/valueset-medication-codes.html

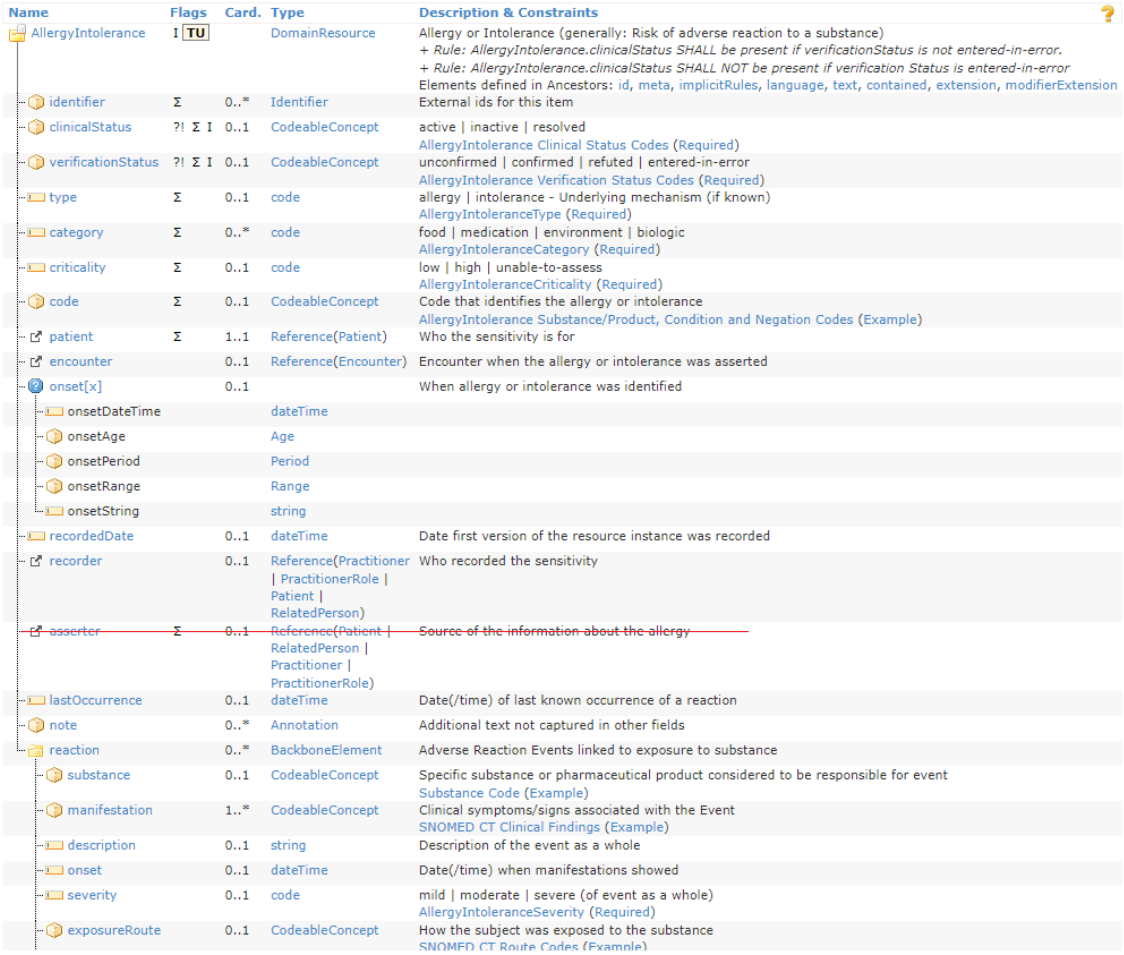
- Immunization history

Name	Flags	Card.	Type	Description & Constraints
Immunization	TU		DomainResource	Immunization event information Elements defined in Ancestors: id, meta, implicitRules, language, text, contained, extension, modifierExtension
identifier		0..*	Identifier	Business identifier
status	?! Σ	1..1	code	completed | entered-in-error | not-done Immunization Status Codes (Required)
statusReason		0..1	CodeableConcept	Reason not done Immunization Status Reason Codes (Example)
vaccineCode	Σ	1..1	CodeableConcept	Vaccine product administered Vaccine Administered Value Set (Example)
patient	Σ	1..1	Reference(Patient)	Who was immunized
encounter		0..1	Reference(Encounter)	Encounter immunization was part of
occurrence[x]	Σ	1..1		Vaccine administration date
occurrenceDateTime			dateTime	
occurrenceString			string	
recorded		0..1	dateTime	When the immunization was first captured in the subject's record
primarySource	Σ	0..1	boolean	Indicates context the data was recorded in
reportOrigin		0..1	CodeableConcept	Indicates the source of a secondarily reported record Immunization Origin Codes (Example)
location		0..1	Reference(Location)	Where immunization occurred
manufacturer		0..1	Reference(Organization)	Vaccine manufacturer
lotNumber		0..1	string	Vaccine lot number
expirationDate		0..1	date	Vaccine expiration date
site		0..1	CodeableConcept	Body site vaccine was administered Codes for Immunization Site of Administration (Example)
route		0..1	CodeableConcept	How vaccine entered body Immunization Route Codes (Example)
doseQuantity		0..1	SimpleQuantity	Amount of vaccine administered
performer	Σ	0..*	BackboneElement	Who performed event
function	Σ	0..1	CodeableConcept	What type of performance was done Immunization Function Codes (Extensible)
actor	Σ	1..1	Reference(Practitioner | PractitionerRole | Organization)	Individual or organization who was performing
note	Σ	0..*	Annotation	Additional immunization notes
reasonCode		0..*	CodeableConcept	Why immunization occurred Immunization Reason Codes (Example)
~~reasonReference~~		~~0..*~~	~~Reference(Condition | Observation |~~	~~Why immunization occurred~~
isSubpotent	?! Σ	0..1	boolean	Dose potency
subpotentReason		0..*	CodeableConcept	Reason for being subpotent Immunization Subpotent Reason (Example)
~~education~~	~~I~~	~~0..*~~	~~BackboneElement~~	~~Educational material presented to patient~~ *+ Rule: One of documentType or reference SHALL be present*
documentType		0..1	string	Educational material document identifier
reference		0..1	uri	Educational material reference pointer
publicationDate		0..1	dateTime	Educational material publication date
presentationDate		0..1	dateTime	Educational material presentation date
programEligibility		0..*	CodeableConcept	Patient eligibility for a vaccination program Immunization Program Eligibility (Example)
fundingSource		0..1	CodeableConcept	Funding source for the vaccine Immunization Funding Source (Example)
reaction		0..*	BackboneElement	Details of a reaction that follows immunization
date		0..1	dateTime	When reaction started
detail		0..1	Reference(Observation)	Additional information on reaction
reported		0..1	boolean	Indicates self-reported reaction
~~protocolApplied~~		~~0..*~~	~~BackboneElement~~	~~Protocol followed by the provider~~
series		0..1	string	Name of vaccine series
authority		0..1	Reference(Organization)	Who is responsible for publishing the recommendations
targetDisease		0..*	CodeableConcept	Vaccine preventatable disease being targetted Immunization Target Disease Codes (Example)
doseNumber[x]		1..1		Dose number within series
doseNumberPositiveInt			positiveInt	
doseNumberString			string	
seriesDoses[x]		0..1		Recommended number of doses for immunity
seriesDosesPositiveInt			positiveInt	
seriesDosesString			string	

Lists

- https://hl7.org/fhir/R4/valueset-vaccine-code.html

- Allergies and intolerances



Name	Flags	Card.	Type	Description & Constraints
AllergyIntolerance	I TU		DomainResource	Allergy or Intolerance (generally: Risk of adverse reaction to a substance) + *Rule: AllergyIntolerance.clinicalStatus SHALL be present if verificationStatus is not entered-in-error.* + *Rule: AllergyIntolerance.clinicalStatus SHALL NOT be present if verification Status is entered-in-error* Elements defined in Ancestors: id, meta, implicitRules, language, text, contained, extension, modifierExtension
identifier	Σ	0..*	Identifier	External ids for this item
clinicalStatus	?! Σ I	0..1	CodeableConcept	active | inactive | resolved AllergyIntolerance Clinical Status Codes (Required)
verificationStatus	?! Σ I	0..1	CodeableConcept	unconfirmed | confirmed | refuted | entered-in-error AllergyIntolerance Verification Status Codes (Required)
type	Σ	0..1	code	allergy | intolerance - Underlying mechanism (if known) AllergyIntoleranceType (Required)
category	Σ	0..*	code	food | medication | environment | biologic AllergyIntoleranceCategory (Required)
criticality	Σ	0..1	code	low | high | unable-to-assess AllergyIntoleranceCriticality (Required)
code	Σ	0..1	CodeableConcept	Code that identifies the allergy or intolerance AllergyIntolerance Substance/Product, Condition and Negation Codes (Example)
patient	Σ	1..1	Reference(Patient)	Who the sensitivity is for
encounter		0..1	Reference(Encounter)	Encounter when the allergy or intolerance was asserted
onset[x]		0..1		When allergy or intolerance was identified
onsetDateTime			dateTime	
onsetAge			Age	
onsetPeriod			Period	
onsetRange			Range	
onsetString			string	
recordedDate		0..1	dateTime	Date first version of the resource instance was recorded
recorder		0..1	Reference(Practitioner | PractitionerRole | Patient | RelatedPerson)	Who recorded the sensitivity
~~asserter~~	~~Σ~~	~~0..1~~	~~Reference(Patient | RelatedPerson | Practitioner | PractitionerRole)~~	~~Source of the information about the allergy~~
lastOccurrence		0..1	dateTime	Date(/time) of last known occurrence of a reaction
note		0..*	Annotation	Additional text not captured in other fields
reaction		0..*	BackboneElement	Adverse Reaction Events linked to exposure to substance
substance		0..1	CodeableConcept	Specific substance or pharmaceutical product considered to be responsible for event Substance Code (Example)
manifestation		1..*	CodeableConcept	Clinical symptoms/signs associated with the Event SNOMED CT Clinical Findings (Example)
description		0..1	string	Description of the event as a whole
onset		0..1	dateTime	Date(/time) when manifestations showed
severity		0..1	code	mild | moderate | severe (of event as a whole) AllergyIntoleranceSeverity (Required)
exposureRoute		0..1	CodeableConcept	How the subject was exposed to the substance SNOMED CT Route Codes (Example)

Lists

- https://hl7.org/fhir/R4/valueset-allergyintolerance-code.html
- https://hl7.org/fhir/R4/valueset-substance-code.html
- https://hl7.org/fhir/R4/valueset-clinical-findings.html
- https://hl7.org/fhir/R4/valueset-route-codes.html

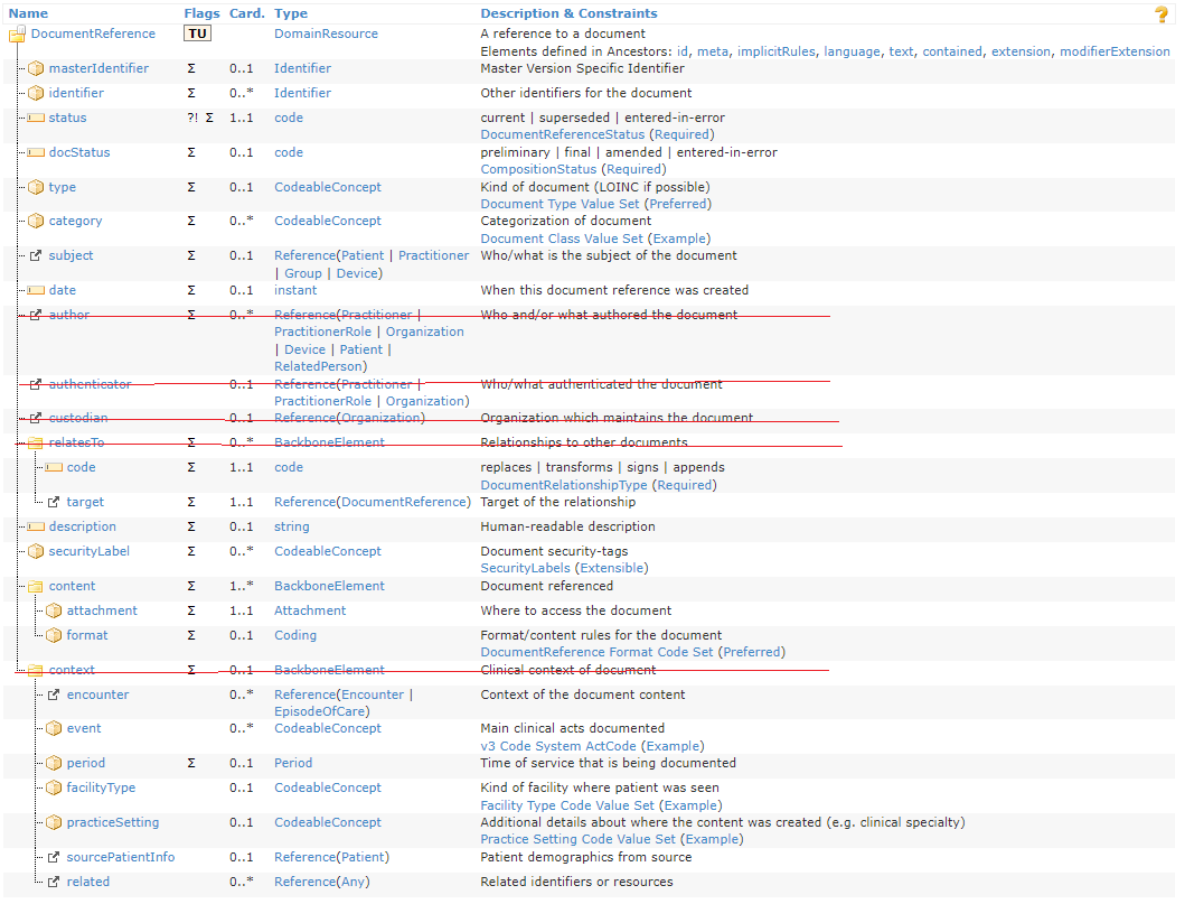
- Vital signs, lab results, and other clinical observations

Name	Flags	Card.	Type	Description & Constraints
Observation	I N		DomainResource	Measurements and simple assertions *+ Rule: dataAbsentReason SHALL only be present if Observation.value[x] is not present* *+ Rule: If Observation.code is the same as an Observation.component.code then the value element associated with the code SHALL NOT be present* Elements defined in Ancestors: id, meta, implicitRules, language, text, contained, extension, modifierExtension
identifier	Σ	0..*	Identifier	Business Identifier for observation
~~basedOn~~	~~Σ~~	~~0..*~~	~~Reference(CarePlan | DeviceRequest | ImmunizationRecommendation | MedicationRequest | NutritionOrder | ServiceRequest)~~	~~Fulfills plan, proposal or order~~
~~partOf~~	~~Σ~~	~~0..*~~	~~Reference(MedicationAdministration | MedicationDispense | MedicationStatement | Procedure | Immunization | ImagingStudy)~~	~~Part of referenced event~~
status	?! Σ	1..1	code	registered | preliminary | final | amended + ObservationStatus (Required)
category		0..*	CodeableConcept	Classification of type of observation Observation Category Codes (Preferred)
code	Σ	1..1	CodeableConcept	Type of observation (code / type) LOINC Codes (Example)
subject	Σ	0..1	Reference(Patient | Group | Device | Location)	Who and/or what the observation is about
~~focus~~	~~Σ TU~~	~~0..*~~	~~Reference(Any)~~	~~What the observation is about, when it is not about the subject of record~~
encounter	Σ	0..1	Reference(Encounter)	Healthcare event during which this observation is made
effective[x]	Σ	0..1		Clinically relevant time/time-period for observation
effectiveDateTime			dateTime	
~~effectivePeriod~~			~~Period~~	
~~effectiveTiming~~			~~Timing~~	
~~effectiveInstant~~			~~instant~~	
issued	Σ	0..1	instant	Date/Time this version was made available
performer	Σ	0..*	Reference(Practitioner | PractitionerRole | Organization | CareTeam | Patient | RelatedPerson)	Who is responsible for the observation
value[x]	Σ I	0..1		Actual result
valueQuantity			Quantity	
valueCodeableConcept			CodeableConcept	
valueString			string	
valueBoolean			boolean	
valueInteger			integer	
valueRange			Range	
valueRatio			Ratio	
valueSampledData			SampledData	
valueTime			time	
valueDateTime			dateTime	
valuePeriod			Period	
dataAbsentReason	I	0..1	CodeableConcept	Why the component result is missing DataAbsentReason (Extensible)
interpretation		0..*	CodeableConcept	High, low, normal, etc. Observation Interpretation Codes (Extensible)
note		0..*	Annotation	Comments about the observation
bodySite		0..1	CodeableConcept	Observed body part SNOMED CT Body Structures (Example)
method		0..1	CodeableConcept	How it was done Observation Methods (Example)
~~specimen~~		~~0..1~~	~~Reference(Specimen)~~	~~Specimen used for this observation~~
~~device~~		~~0..1~~	~~Reference(Device | DeviceMetric)~~	~~(Measurement) Device~~
~~referenceRange~~	~~I~~	~~0..*~~	~~BackboneElement~~	~~Provides guide for interpretation~~ *+ Rule: Must have at least a low or a high or text*
low	I	0..1	SimpleQuantity	Low Range, if relevant
high	I	0..1	SimpleQuantity	High Range, if relevant
type		0..1	CodeableConcept	Reference range qualifier Observation Reference Range Meaning Codes (Preferred)
appliesTo		0..*	CodeableConcept	Reference range population Observation Reference Range Applies To Codes (Example)
age		0..1	Range	Applicable age range, if relevant
text		0..1	string	Text based reference range in an observation
~~hasMember~~	~~Σ~~	~~0..*~~	~~Reference(Observation | QuestionnaireResponse | MolecularSequence)~~	~~Related resource that belongs to the Observation group~~
~~derivedFrom~~	~~Σ~~	~~0..*~~	~~Reference(DocumentReference | ImagingStudy | Media | QuestionnaireResponse | Observation | MolecularSequence)~~	~~Related measurements the observation is made from~~
~~component~~	~~Σ~~	~~0..*~~	~~BackboneElement~~	~~Component results~~
code	Σ	1..1	CodeableConcept	Type of component observation (code / type) LOINC Codes (Example)
value[x]	Σ	0..1		Actual component result
valueQuantity			Quantity	
valueCodeableConcept			CodeableConcept	
valueString			string	
valueBoolean			boolean	
valueInteger			integer	
valueRange			Range	
valueRatio			Ratio	
valueSampledData			SampledData	
valueTime			time	
valueDateTime			dateTime	
valuePeriod			Period	
dataAbsentReason	I	0..1	CodeableConcept	Why the component result is missing DataAbsentReason (Extensible)
interpretation		0..*	CodeableConcept	High, low, normal, etc. Observation Interpretation Codes (Extensible)
referenceRange		0..*	see referenceRange	Provides guide for interpretation of component result

Lists

- https://hl7.org/fhir/R4/valueset-observation-category.html
- https://hl7.org/fhir/R4/valueset-observation-codes.html
- https://hl7.org/fhir/R4/valueset-body-site.html
- https://hl7.org/fhir/R4/valueset-observation-methods.html
- https://hl7.org/fhir/R4/valueset-observation-codes.html

- Clinical notes & other patient documents



Name	Flags	Card.	Type	Description & Constraints
DocumentReference	TU		DomainResource	A reference to a document Elements defined in Ancestors: id, meta, implicitRules, language, text, contained, extension, modifierExtension
masterIdentifier	Σ	0..1	Identifier	Master Version Specific Identifier
identifier	Σ	0..*	Identifier	Other identifiers for the document
status	?! Σ	1..1	code	current | superseded | entered-in-error DocumentReferenceStatus (Required)
docStatus	Σ	0..1	code	preliminary | final | amended | entered-in-error CompositionStatus (Required)
type	Σ	0..1	CodeableConcept	Kind of document (LOINC if possible) Document Type Value Set (Preferred)
category	Σ	0..*	CodeableConcept	Categorization of document Document Class Value Set (Example)
subject	Σ	0..1	Reference(Patient | Practitioner | Group | Device)	Who/what is the subject of the document
date	Σ	0..1	instant	When this document reference was created
author	Σ	0..*	Reference(Practitioner | PractitionerRole | Organization | Device | Patient | RelatedPerson)	Who and/or what authored the document
authenticator		0..1	Reference(Practitioner | PractitionerRole | Organization)	Who/what authenticated the document
custodian		0..1	Reference(Organization)	Organization which maintains the document
relatesTo	Σ	0..*	BackboneElement	Relationships to other documents
code	Σ	1..1	code	replaces | transforms | signs | appends DocumentRelationshipType (Required)
target	Σ	1..1	Reference(DocumentReference)	Target of the relationship
description	Σ	0..1	string	Human-readable description
securityLabel	Σ	0..*	CodeableConcept	Document security-tags SecurityLabels (Extensible)
content	Σ	1..*	BackboneElement	Document referenced
attachment	Σ	1..1	Attachment	Where to access the document
format	Σ	0..1	Coding	Format/content rules for the document DocumentReference Format Code Set (Preferred)
context	Σ	0..1	BackboneElement	Clinical context of document
encounter		0..*	Reference(Encounter | EpisodeOfCare)	Context of the document content
event		0..*	CodeableConcept	Main clinical acts documented v3 Code System ActCode (Example)
period	Σ	0..1	Period	Time of service that is being documented
facilityType		0..1	CodeableConcept	Kind of facility where patient was seen Facility Type Code Value Set (Example)
practiceSetting		0..1	CodeableConcept	Additional details about where the content was created (e.g. clinical specialty) Practice Setting Code Value Set (Example)
sourcePatientInfo		0..1	Reference(Patient)	Patient demographics from source
related		0..*	Reference(Any)	Related identifiers or resources

Lists

- https://hl7.org/fhir/R4/valueset-c80-doc-typecodes.html
- https://hl7.org/fhir/R4/valueset-document-classcodes.html
- https://hl7.org/fhir/R4/valueset-formatcodes.html

Future Plans

As a developer, I am competent at developing the applications and do not have the other expertise required in order for the project to be a success. I therefore require funding and assistance from industry professionals.

1. Funding
   This is required for hosting the system online and immediately enabling its use by interested doctors, facilities, patients etc. Approximate costs per month are E 300.00.
2. Marketing
   The system needs to be marketed to the correct clients that may require the services of having their patient record available to them whenever.
3. Stakeholder Engagement
   Stakeholders, mostly health facilities and doctors need to be engaged for them to be aware of the product and on-board their patients. They can also make further requirements for functionality to be added to the system
4. Business Partners
   In order for the people actively applying efforts towards the success of the project to be driven all the time, it would be best for them to own a percentage of the product and this shall be negotiated.